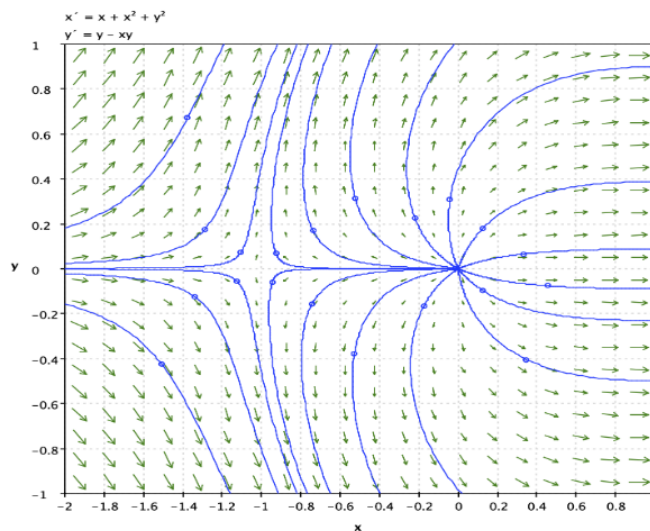
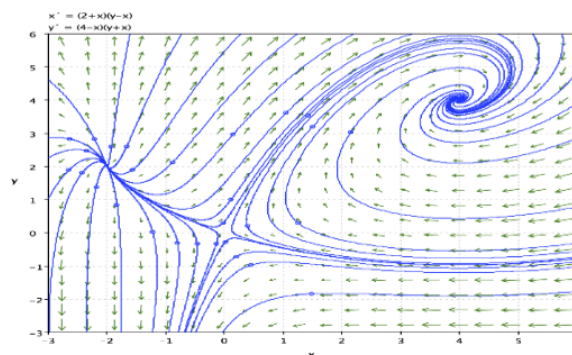


(10 points) *Problem 2.* Consider a planar nonlinear ODE system $\begin{cases} x'(t) = x + x^2 + y^2, \\ y'(t) = y - xy. \end{cases}$

- (i) Find its **two equilibria** (by either solving algebraic equations or examining the given graphics).
- (ii) For each equilibrium, calculate its Jacobian matrix and the eigenvalues. Then determine its type (saddle-point, nodal source, center, etc.) and stability.



(10 points) *Problem 2.* Consider the ODE system $\begin{cases} x'(t) = (2+x)(y-x) \\ y'(t) = (4-x)(y+x) \end{cases}$ and the following figure.



Answer the following questions.

- (i) Find the coordinates of the 3 equilibria (Solving the algebraic system to get the full credit).
- (ii) For each equilibrium, find the Jacobian matrix and its eigenvalues, determine its type:
Nodal source Nodal sink Spiral source Spiral sink Saddle-point
- (iii) Identify stability for each equilibrium:
Unstable Asymptotically stable Stable but not asymptotically stable